Individual Weekly Report – Week 1

Team Formation & Dataset Selection

Name	Madhumitha S
Roll No	25BAD053
Department	Artificial Intelligence and Data Science
Project Title	Youth Smoking and Drug Consumption: From Raw Data to Visual Insights
Role	Data Visualization and Dashboard Development
Week	Week 1 – Team Formation & Dataset Selection

Objective of the Week

The aim of the first week was to form the team, select the dataset, and decide how the project results would be shown visually. My responsibility was the visualization side of the project, so I started preparing for it from the very beginning.

Work Done

Researching dashboard tools: I looked at different tools that can be used to build a dashboard, such as Streamlit and Plotly. I compared them based on how easy they are to learn, whether they support interactive charts, and whether they suit a Python-based data science project.

Exploring visualization techniques: I studied the common chart types, such as bar charts, line charts, histograms, boxplots, scatterplots and pie charts. I read about when each chart should be used, so that I could choose the right one for each kind of data later in the project.

Studying how data can be presented: I learned how charts and graphs make large data easier to understand. I noted important points such as using clear titles, proper labels, readable colours and simple layouts so that the message of each chart is clear.

Discussing ideas with the team: I shared my visualization ideas with my team members and listened to their suggestions about the dataset. This helped us make sure that the visualization plan matched the project topic of youth smoking and drug consumption.

Tools and Techniques Used

Research on Streamlit and Plotly, study of basic chart types, and team discussions.

Outcome of the Week

By the end of the week, I had a clear idea of which tools and chart types could be used in the project. The team also had a common understanding of how the final results should be presented.

What I Learned

I learned that choosing the right tool and chart type at the beginning saves time later. I also understood that discussing ideas early with the team avoids confusion in the later stages.

Contribution to the Project

My research gave the team a starting direction for the visual part of the project and prepared me for the data understanding work of the next week.

Challenges Faced

Choosing between several dashboard tools and chart types was not easy, because each option has its own advantages. I handled this by comparing them on simple points such as ease of use and interactivity, and by taking the team's opinion before deciding.

Plan for Next Week

In Week 2, I will study the structure and variables of the selected dataset, so that I can match them with suitable visualizations and plan the dashboard layout.

Weekly Summary

Week 1 gave me the base for my role. By researching tools and techniques and discussing ideas with the team, I was ready to start working with the actual data.